

A machine is a combination of rigid or resistant bodies having a definite motion & capable of performing a useful work.

Machines differ in appearance, function, and complexity from the simple hand-operated tools to the aeroplane or ocean liner.

No matter how complicated or big a machine is, it may be broken into smaller assemblies until an analysis of the parts operation becomes dependent upon an understanding of a few basic concepts. (the concept are mostly from the elementary physics)

A Mechanism — is that part of a machine which contains two or more pieces so arranged that the motion of one compels the motion of the others, all in fashion prescribed by the nature of the combination.

Mechanism applies only to the physical arrangement that provides for the motions of definite motion of the parts of a machine. e.g. a wristwatch is a mechanism, but it does no useful work.

Mechanisms are mechanical means for conversion of motion, transmission of power or the control of these.

Mechanisms are not always limited to mechanical means in the modern days, they may include pneumatics, hydraulics, electrical and electronic elements.

Mechanisms are found in Internal Combustion Engine, Compressors, locomotives, agricultural, earthmoving equipment, excavating, mining, packaging, textile, & so many other machineries (machine tools, printing presses, engraving m/c, transmission, washing m/c, land mowers, sewing m/c, projectors, toys, door locks, counters, microswitches, speedometer etc.)
* Name any ten ~~types~~ m/cs that contain mechanisms

Machine elements are the elementary mechanical parts used as building blocks for the construction of most devices, apparatus, and machinery. The most common of the m/c elements are the gears, the fundamentally ~~is~~ is a combination of the wheel & the lever to form a toothed wheel. The rotation of this gear on a hub or shaft drives other gears which may rotate faster or slower, ~~depend~~ depending upon the no of teeth on the basic wheels.

A wheel must have a shaft on which it may rotate. The wheel is fastened to the shaft with a key, & the shaft is joined to other shaft with couplings. The shaft must rest in a bearing such as journal bearings, ball bearings, or roller bearings.

The shaft may be started by a clutch or stopped by a brake, it may be

Turned by a pulley with a flat or V-belt, or a chain or a rope ~~connected~~ connecting ~~the~~ it to the pulley on a second shaft.

The supporting structure may be built or assembled with bolts, ~~rope~~ rivets or welding. proper application of the elements depends upon the knowledge of the forces on the structure & the ^{strength} ~~material~~ of the materials employed.

2.0 DESIGN:

(P1)

Design is the process of creating a new and better machines and ~~improving~~ an existing machine.

A ~~better~~ new better machine means, a machine that is more economical in overall cost of production and operation. Design is a communication intensive activity in which both words & pictures are used, written & oral forms are employed. Engineers have to communicate ^{effectively} ~~effectively~~ and work with people of many ^{backgrounds} ~~backgrounds~~.

A designer's personal resources of Creativeness, Communication ability & Problem Solving skill are intertwined with knowledge of Technology & first principles. Engineering tools such as mathematics, Statistics, Computers, Graphics & Languages are combined to produce a plan that

when carried out produces a product that is functional, safe, reliable, competitive, usable, manufacturable & marketable, regardless of ~~which~~ who builds it or who uses it.

2.1 Classification of M/c design

M/c design are classified as follows.

2-19
a)

Adaptive design — The work here is concerned with the adaptation of ~~the~~ ^{an} existing work or design; it does not need special knowledge or skills & can be adapted by designers of ordinary technical training. The designer makes minor alterations or modifications.

2-16.
b.

Development design — This type needs considerable scientific training & design ability in order to modify the existing design into ~~a~~ new idea by adopting

a new material or different method (2)
of manufacturing process. In this case
the final product may differ ^{quite} from
markedly from the original product.

2.16. New design — This type needs
a lot of research, Technical ability &
Creative thinking, This type is only taken
by those with personal qualities of a
sufficiently high order.

2.2. CLASSIFICATION OF DESIGN

B-1 methods

2.2a. Rational design — This depends
on mathematical formulae of principles
of mechanics.

2.2b. Empirical design — This type depends
upon the ^{Empirical} ~~production~~ formulae ~~based~~ on

The practice & past experience.

2.2c. Industrial design - This type depends on the production aspects to manufacture any m/c component in the industry.

2.2d. Optimum design - its the best design for the given objective function under the specified constraints. It may be achieved by minimising the undesirable effects.

2.2e. System design - design of any complex mechanical system like a motor car.

2.2f. Element design - design of any element of the mechanical system like pistons, crank shaft, connecting rods etc.

2.2g. Computer Aided design - depends on the use of Computer to assist in creation

Modifications, analysis & optimization. (3)

3. MACHINE DESIGN CONSIDERATIONS

1. Type of load — The load on a m/c component, may act in several ways — due to which internal stresses are set up.
2. Motion of parts — Successful operation of any machine depends largely upon the internal stress arrangement of the parts which will give the motion required.
3. Material Selection — Different materials withstand different amount of force/stress, thus a proper material has to be selected for a certain m/c or m/c parts.
4. Form & Size of parts — It is necessary to know the forces which the parts must sustain.

5. Frictional resistance and lubrication -

There is always a loss of power due to frictional resistance & it should be noted that the friction of starting is higher than that of running. Therefore, attention must be given to lubrication of parts.

6. Ease of manufacturing & assembling of parts

7. Ease of service & replacement of parts

8. Safety of operation - The designer has to provide safety devices / means for the operator of any machine.

9. Cost - Cost of construction must be at the least minimum which can be acceptable to the community, b/cos, if the cost of a material is high, it can be a further consideration